

Feriel Yala

PHD STUDENT

Sherbrooke, Qc, Canada

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Final-year PhD candidate in Bioinformatics and Biochemistry with a Master's in In-Silico Drug Design (France). My key strength is leading end-to-end projects, coordinating a team, and translating complex technical concepts into clear, actionable insights for non-technical stakeholders.

Experiences

PhD in Bioinformatics - Biochemistry

Sherbrooke, Canada

XAVIER ROUCOU LAB - DEPARTMENT OF BIOCHEMISTRY AND FUNCTIONAL GENOMICS, UNIVERSITY OF SHERBROOKE

Since September 2022

Project : Large-scale structural and functional annotation of microproteins.

PhD centered on large-scale prediction, clustering, and functional annotation of non-canonical microproteins, combining structural bioinformatics, HPC workflows, and integrative data analysis.

- Contributed to, maintained, and exploited the OpenProt proteogenomic database, working with large-scale structural and functional protein data.
- Performed high-throughput clustering and alignment on more than 53 million predicted protein structures.
- Built and ran large-scale computational pipelines using Linux, Python, PostgreSQL, SQLite, and HPC environments (Digital Research Alliance of Canada).
- Published as first author in a peer-reviewed journal; second manuscript currently in preparation.
- Delivered four departmental research seminars and presented multiple talks and posters at national and international conferences.
- I also collaborated with several research groups by providing bioinformatics support and assisting students from different laboratories, allowing me to manage multiple projects simultaneously and develop a strong sense of priority management.
- I troubleshoot unexpected results and handled system failures or HPC outages while managing stress and maintaining focus and problem-solving efficiency.

Research Intern

Sherbrooke, Canada

XAVIER ROUCOU LAB - UNIVERSITY OF SHERBROOKE

Jan. 2022 – Jun. 2022

Project : Designed and implemented a large-scale structural prediction pipeline for 900,000+ non-canonical microproteins integrated into a proteogenomics database OpenProt. Configured and optimized MMseqs2 on high-performance computing infrastructure (Digital Research Alliance of Canada), Deployed AlphaFold2 at scale using Apptainer/Singularity containers on HPC, Automated and orchestrated large-scale computational workflows and Integrated validated structural predictions into the OpenProt database.

Acted as a bridge between computational and experimental teams: clear communication of bioinformatics concepts, leading discussions, and presenting visual summaries for wet-lab researchers.

Research Intern

Paris, France

INSERM

Apr. 2021 – Jul. 2021

Project: Identification of inhibitors targeting the SARS-CoV-2 Spike protein.

Contributed to large-scale virtual drug repurposing efforts during the SARS-CoV-2 pandemic, Performed high-throughput virtual screening of FDA-approved molecules targeting the Spike-ACE2 interface on a high-performance GPU cluster, Used Python and R for data processing, visualization, docking analysis, ranking, and workflow automation.

Analyzed protein-ligand interactions at the structural level to identify promising candidates, strengthened practical skills in structure-based drug design and computational screening at scale.

Strategy Analyst

Paris, France

HARMONIE MUTUELLE

Jan. 2019 – May. 2019

Project : How can Harmonie Mutuelle leverage regulatory developments in the healthcare sector to create a commercial advantage?

Analyzed customer profiles to identify high-potential prospect segments, Proposed fast and cost-efficient acquisition strategies to increase membership, Developed strong analytical autonomy and rapid learning skills, Adapted quickly to a non-scientific, business-oriented environment.

Built understanding of organizational, economic and customer-driven constraints.

Translated analyses into practical, actionable recommendations for decision-makers.

Demonstrated ability to integrate into new environments and contribute effectively outside my scientific background.

Education

PhD in Bioinformatics - Biochemistry

Sherbrooke, QC, Canada

DEPARTMENT OF BIOCHEMISTRY AND FUNCTIONAL GENOMICS, UNIVERSITY OF SHERBROOKE

Since September 2022

PhD focused on the structural and functional characterization of non-canonical microproteins using large-scale bioinformatics pipelines.

Key modules: Python & PostgreSQL, SQLite, Advanced Statistics, Data Analysis, Databases, Big Data, Unix, structural biology and biocomputing, multi-omics, scientific communication and seminar & HPC.

Master's degree - Bioinformatics - In Silico Drug Design (ISDD)

Paris, France

UNIVERSITY PARIS CITE

Sep. 2020 - Aug. 2022

Key modules: Machine Learning, Databases, Programming in Python and R, Biostatistics and data analysis, Understanding of therapeutic targets, protein structure, folding, and interactions, Molecular docking and virtual screening.

Bachelor's Degree – Biochemistry, Integrative Biology and Physiology

Paris, France

UNIVERSITY PARIS CITE

Sep. 2018 – Aug. 2020

Key modules: Biochemistry, biotechnology, Biostatistics, quality control principles and laboratory practices, microbiology, virology, and immunology, molecular and cellular biology, enzymology.

Bachelor's Degree – Microbiology

Algiers, Algeria

UNIVERSITY OF SCIENCE AND TECHNOLOGY HOUARI BOUMEDIENE

Sep. 2015 – Aug. 2028

Key modules: Biochemistry, biotechnology, Biostatistics, quality control principles and laboratory practices, microbiology, virology, and immunology, molecular and cellular biology, enzymology.

Languages

French Native

English Full professional proficiency

Technologies

Langages Python (Numpy, Pandas, Matplotlib, Seaborn, Plotly), R

Database SQL, SQLite, PostgreSQL

Cloud & Data Platforms Microsoft Azure, Databricks

DevOps Apptainer, Docker

Version Control Git

Skills

Operating system Unix/Linux environments

Data processing & programming

Data processing, visualization, ML, automation, R, High-performance computing, Workflow automation & Large-scale pipelines

Data science for biology

Machine learning for molecular/biological data , Biostatistics & data analysis, Graphical and statistical visualization

Computational biology

Multi-omics & Integrative Analysis, Familiar with major drug and compound databases (DrugBank, ZINC), Large-scale molecular docking, High-throughput virtual screening

Soft skills & communication

Scientific communication, Adaptability to new environments, Analytical autonomy & fast learning, Teamwork & cross-functional collaboration, Sense of responsibility & project ownership, Ability to translate analysis into actionable insights

Scientific communication

Nov. 2025	Human Proteome Organization (HUPO) , Structure-based clustering reveals functional insights into microproteins encoded by non-canonical open reading frames	Toronto, Canada
May 2025	Proteo , Beyond sequences: advancing the study of microproteins through structural clustering	Sherbrooke, Canada
Jul. 2024	International Society for Computational Biology (ISCB/ISMB) , Docking between proteins expressed from the same mRNA to test for direct functional cooperation	Montreal, Canada
May 2024	Proteo , Docking between proteins expressed from the same mRNA to test for direct functional cooperation	Montreal, Canada
May 2023	Biology of genomes , Predicting the Structure and Function of Alternative Proteins	New-York, USA
May 2022	Journee Phare , OpenProt, version3: addition of structures and intrinsic disorder to accelerate research on alternative proteins	Val-des-Sources, Canada

Publications

OpenProt 2.0 builds a path to the functional characterization of alternative proteins. *Nucleic Acids Research*
CO-FIRST AUTHOR *Jan. 2024*

- OpenProt is an open-access proteogenomic resource that catalogs non-canonical open reading frames and their associated proteins across multiple species. It enables large-scale functional annotation and analysis, supporting the systematic exploration of previously unannotated proteins and the cryptic proteome.

Structure-based clustering reveals functional insights into microproteins and larger proteins encoded by non-canonical open reading frames.
FIRST AUTHOR *2025*

- Article in preparation.